

Sets

Coin toss : $\{\text{heads}, \text{tails}\}$

Element of set

set

Dice roll : $\{1, 2, 3, 4, 5, 6\}$

Since we assume the above sets represent all possible outcomes within their respective contexts, we can call these universal sets (S).

Denoted "Sample Space"

Associativity: $A \cup B \cup C = A \cup (B \cup C) = (A \cup B) \cup C$

Distributivity: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$